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Fault Detection and Classification in Transmission Lines using Artificial Intelligence

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***Abstract*—Large interconnected electric power transmission lines are vital for delivery of continuous power flow to various loads. Due to this wide spread phenomena, transmission lines are more prone to electric faults and may lead to poor reliability of the power system. Thus, this paper tries to detect and classify various types of faults on interconnected transmission lines by utilizing artificial intelligence. A Back Propagation Neural Network (BPNN) is developed with three phase voltages/currents with zero sequence components. The performance of proposed BPNN based fault identification methodology is validated under various symmetrical and unsymmetrical faults on the IEEE 14-bus test power system. The simulation results reveal that the proposed fault identification methodology has accurately detected and classified all types of transmission line fault scenarios in the complex power system. Thus, the proposed artificial intelligence based fault identification concept can be quite suitable for protection of transmission lines in the electric grid.**

***Index Terms*—Artificial intelligence, Back propagation neural network, Fault identification, Levenberg-Marquardt algorithm, Transmission line protection**

I. Introduction

Generally, the electrical power system comprises of generation, transmission and distribution networks with complex interconnections. Long and large number of transmission lines interconnect the remotely located power generating plants and distributed loads. Due to above scenarios, power transmission lines are exposed to the environment and are vulnerable to various challenges specifically towards electric faults [1]. The electric fault is an abnormal condition that occurs between two phases or between one phase and ground in a transmission line. The voltage at the faulted point drops and causes overheating due to large current flows and eventually damages the large parts of the power system equipments.

Usually, symmetrical faults and unsymmetrical faults, which are collectively known as shunt faults, frequently occur on complex interconnected transmission lines [2]. Single line-to-ground faults, Line-to-line faults and double line-to-ground faults are the three types of unsymmetrical faults. Single line-to-ground faults, which account for 70% of transmission line defects, are the most common. Further, the Line-to-line faults account for 15% of all faults, whereas double line-to-ground faults account for 10% only. The remaining 5% of the faults are classified as three-phase faults class or symmetrical faults [1]. Triple line (L-L-L) and Triple line-to-ground (L-L-L-G) are the two types of symmetrical faults.

A robust fault identification scheme should provide an effective, reliable, fast and secure way of relaying operation to remove the electric fault through circuit breakers. Traditional class of fault identification methodologies make use of electro-mechanical, static and numerical relays which mechanise the relays to send trip signals to the associated circuit breakers in the event of abnormalities i.e. faults in the power system. Such protection schemes are may turn to be quite costly due to sophisticated configuration of electronic components required for adaptive relaying process.

Various researchers have suggested solutions by using Discrete Wavelet Transform (DWT) [3], Fast Fourier Transform (FFT) [4-5], fuzzy logic [6] and adaptive Neuro-Fuzzy Inference System (ANFIS) [7], etc. [3] has proposed a DWT technique wherein the sampling of voltages and currents at both ends of the transmission lines by using Global Positioning System (GPS) but suffers from shift sensitivity and directionality. In [4-5], the harmonics of fault information is utilized to detect the faults through travelling wave theory. A fuzzy/neuro logic based on-line fault detection and classification of fault in lines is demonstrated in [6-7] which is very sensitive to inaccurate inputs; needs tedious training of network.

In literature, various methodologies are suggested for electric fault detection and classification in power lines by using artificial neural networks (ANN). But, most of suggested strategies are trying to detect and classify the unsymmetrical class of electric faults only. In this paper, the authors have tried to implement the detection and classification models for both symmetrical and unsymmetrical electric faults related to interconnected transmission lines by utilizing artificial intelligence i.e. ANN with Levenberg-Marquardt algorithm [9].

The remaining paper is organized as Section II describes the proposed ANN methodology for electric fault detection and classification in interconnected power transmission lines with inherent attribute of pattern recognition. The validation of proposed ANN methodology for electric fault identifications under various dynamic scenarios is explained in Section III. Section IV concludes the work of the paper.

II. Proposed ANN based Fault Identification

Artificial Neural Networks (ANN) are computational models that mimic the way nerve cells work in the human brain. ANN has its roots in the nervous system of the human body and the human brain and also, the neuroscience associated with it, which allows humankind to think and reason against various scenarios. ANN is an adaptive and non-linear system that can interpret and implement any physical phenomena by feeding the associated past data into it. Further, the ANN analyses the data fed into it, creates and recognizes patterns,and implements the patterns generated into the new data given to it to analyse.

Thus, the authors have inspired the above artificial machine learning process and tried to mimic for identification of electric faults in the interconnected power system. The detailed theory and proposed ANN based fault identification mechanism is described in the subsequent subsections.

A. *Theory of artificial neural network (ANN)*

Artificial Neural Network (ANN) is an interconnected group of artificial neurons. It replicates the neural network of a human brain and the functioning of the human brain as well. ANN has three standard layers i.e. input layer, hidden layer(s), and output layer. One or more hidden layers are connected to the input layer to realize the accurate physical phenomena behaviour i.e. attainment of prescribed target values. The neurons in the hidden layer are known as perceptrons which are inter-linked to the output layer.

The ANN can be realised by employing the two learning algorithms i.e. feed forward and back propagation algorithms. The input data in the feed forward method moves forward from the input nodes to the output nodes via hidden nodes. whereas, the backward propagation is the process of travelling backward from the output to the input layer. The error values are randomly assigned as weights, and biases are reduced in order to produce the correct output. The detailed concept of feed forward and back propagation algorithms is explained in the subsequent sections.

1) Feed forward neural network: Artificial neurons are the fundamental building blocks of an artificial neural network. In mathematical form, such network models allow to solve various challenges in science and engineering domains. Usually, artificial neural networks utilize an activation function to determine whether or not a neuron should be engaged in the process of signal generation to another neuron.

The binary step activation function, $f(x)$ of a node determines the output of the respective node for given an input or group of inputs which is mathematically expressed as:

$$f(x) = \begin{cases} 1, x \geqslant \Theta \\ 0, x < \Theta \end{cases} \tag{1}$$

where, x is input e.g. voltage/current/sequence components, Θ is threshold limit to be reached or extended for the neuron to produce a sufficient signal, f is a linear or a nonlinear function that acts on the produced signal.

When the input surpasses threshold limit (Θ), the function $f(x)$ returns 1 (or true), but when the input does not reach the threshold limit (Θ), it returns 0.

The basic neuron model transfer function can be mathematically expressed as:

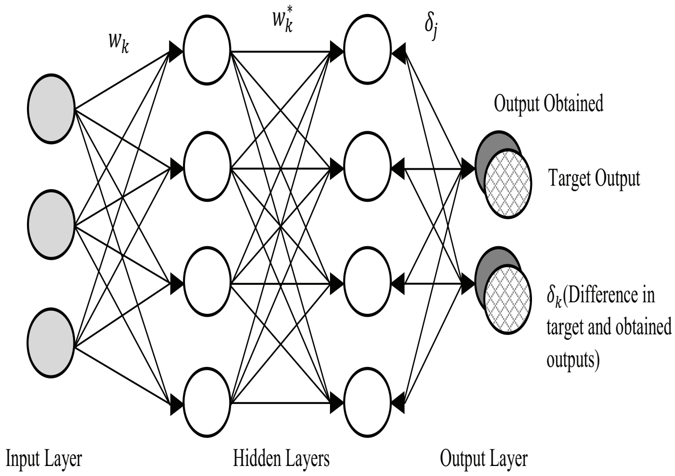

Fig. 1. The general structure of BPNN

$$y = f(net) \tag{2}$$

where, y is output of neuron of a concerned layer i.e hidden/output layers, $net = \sum_{i=0}^{n} x_i w_i$, n is number of neurons in the input layer, w_1, w_2, w_3,.....w_n are associated weights of edges between neurons.

The conditions for each neuron firing in the layers are mathematically expressed as:

$$\sum_{i=0}^{n} x_i w_i \geqslant \Theta \tag{3}$$

$$f(net) \geqslant \Theta \tag{4}$$

2) Back propagation neural network (BPNN): The accuracy of modeling of the physical phenomena can be improved through Back-propagation method by computation of the gradient for nonlinear multi-layer neural network. In back-propagation process, the output neuron is used to evaluate the correction required in predefined weight values. Thus, the BPNN weights are randomly selected as well returns with a pair of input as shown in the given Fig. 1. The detailed concept of BPNN is explained below:

The output of a neuron at j^{th} iteration i.e. (y_j) in a layer is mathematically expressed as

$$y_j = \sum_{i=0}^{n} w_{kj}^{(2)} z_j \tag{5}$$

where, $z_j = f(a_j)$, $a_j = \sum_{i=0}^{n} w_{ji}^{(1)} x_i$, a_j is inputs' weighted sum, w_{ji} is amount of weight that the edges carry, x_i is inputs e.g. voltage/current/sequence components, z_j is activation unit that sends a link to neuron at j^{th} iteration in the layer.

The error derivative i.e. δ_k at the k^{th} neuron is computed as

$$\delta_k = y_k - t_k \tag{6}$$

where, y_k is k^{th} neuron activation function output, t_k is input's associated target.

Further, the derivative error with respect to a_j in the hidden layers can be computed through back propagation method which is mathematically expressed as

$$\delta_j = (1 - z_j^2) \sum_{k=1}^{k} w_{kj} \delta_k \quad (7)$$

Moreover, the weights are updated/restructured to a new value by utilizing the following expression

$$w_{ji}{}^{*} = w_{ji} - \frac{\partial \delta_k}{\partial w_{ji}} \quad (8)$$

where, w_{ji}^{*} is updated weights of edges after back-propagation.

The above process is to be repeated for all available training data set of inputs-outputs arrangements and network meets the given target values with predefined amount of error tolerance i.e. Mean square error (MSE). The MSE for a Back Propagation Neural Network (BPNN) can be mathematically computed as

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \widehat{Y}_i)^2 \quad (9)$$

where, n is number of data inputs,Y_i is observed values, $\widehat{Y}_i$ is predicted values.

In this paper, n is the total number of observations i.e. the three phase voltages and currents and also the sequence components that comprise of data-set i.e. (V_a, V_b, V_c, I_a, I_b, I_c, V_z and I_z). The observed values (Y_i) are the values that related to fault voltages, currents, and sequence components and the predicted values or Target Values i.e. $(\widehat{Y}_i)$ are the values of three phase voltages, currents, and sequence components when no fault occurs. Whenever a deviation takes place from the target values, it indicates that a fault has occurred and same has been trained through artificial intelligence i.e. neural network.

B. Methodology for fault identification

By utilizing above discussed artificial intelligence neural network, authors tried to construct the Back Propagation Neural Network (BPNN) for detection and classification of various electric faults in the large interconnected power system. On first, various electrical fault scenarios i.e. symmetrical and unsymmetrical shunt faults including no fault scenario are simulated on standard test power system to generate the associated data-sets.

The generated data-sets are utilized to train the Back Propagation Neural Network (BPNN) MATLAB toolbox through Levenberg-Marquardt algorithm [9] for effective fault detection and classification under various abnormal scenarios in the power system. The flow diagram to generate data-sets and training of BPNN is depicted in Fig. 2.

Among available data-set, only 70% of the data-set is utilized for training the Back Propagation Neural Network (BPNN) to accurately model the proposed fault identification strategy i.e. computation of appropriate weights. Thus, the trained neural network model can recognize the pattern for a given input of voltage/current/sequence components and identify the faulty scenario in the power system.

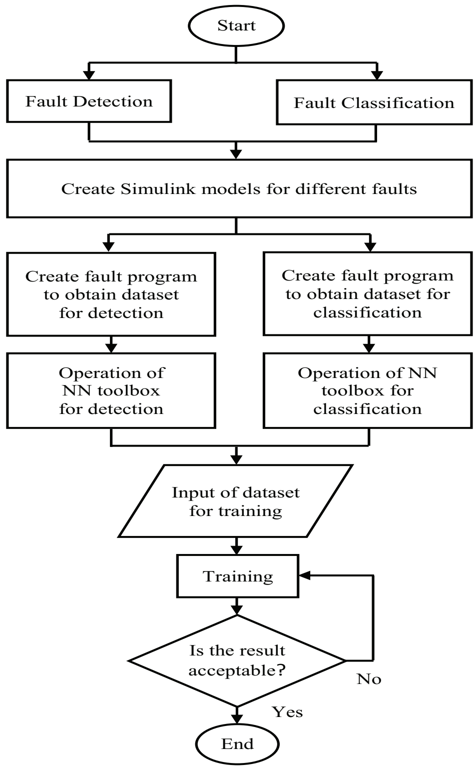

Fig. 2. Proposed fault detection and classification strategy using BPNN

The validation of BPNN is a process in which a portion of entire data-set is used to check how efficiently the predictive model work, failing which the neural network blocks are retrained. Thus, only 15% of the entire data-set is utilized for validation of the trained neural network to recheck the fault identification performance under different abnormal scenarios.

Finally, a portion of the data-set i.e. remaining 15% of data is utilized to test the validated BPNN. Based on the overall efficiency and accuracy of the constructed BPNN model, the artificial intelligence predictive model can be suitable for identification electrical faults in the complex interconnected power system. The proposed artificial intelligence i.e. Back Propagation Neural Network (BPNN) based fault detection and classification mechanism for protection of transmission lines is depicted in Fig. 2.

The performance of above proposed artificial intelligence strategy for fault identification in interconnected transmission lines is described in the next section.

III. Simulation Results

An IEEE 14-bus test power system [8] is considered to showcase the performance of proposed fault detection and classification in transmission lines through artificial intelligence i.e. Back Propagation Neural Network (BPNN). The above test power system comprises of 14 buses, 5 generators, and 11 loads, as shown in Fig. 3. The IEEE 14-bus test system

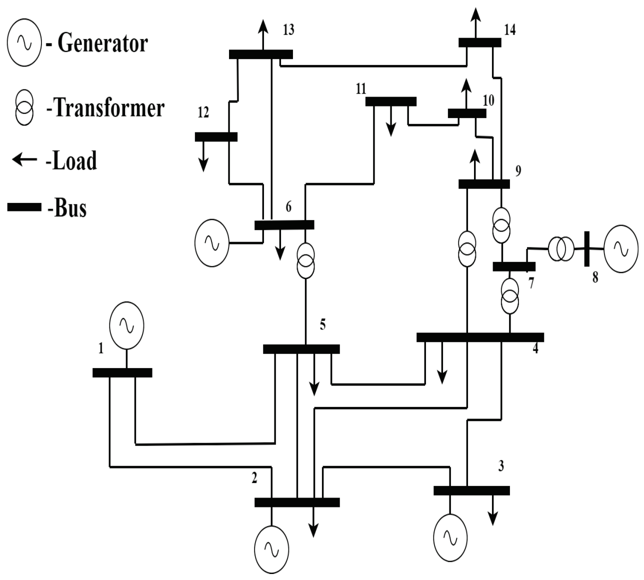

Fig. 3. Single line diagram of IEEE 14-bus test power system

is simulated on MATLAB software under different fault scenarios which includes nine different types of unsymmetrical fault conditions, including line-to-ground (A-G, B-G, C-G), line-to-line (A-B, B-C, A-C), double line-to-ground (A-B-G, B-C-G, A-C-G), and no fault. It also includes symmetrical fault (A-B-C-G, A-B-C).

TABLE I
TARGET OF BPNN OUTPUT LAYER FOR FAULT DETECTION

Fault Type	Network Output
No Fault	0
A-G	1
B-G	1
C-G	1
A-B-G	1
A-C-G	1
B-C-G	1
A-B	1
A-C	1
B-C	1
A-B-C	1
A-B-C-G	1

A. Validation of proposed BPNN model

The performance of proposed transmission line fault identification mechanism using artificial intelligence (BPNN) is revealed by building the appropriate BPNN with minimum Mean square error (MSE) and accurate regression characters tics during fault detection and classification under low and high resistance faults on the transmission lines.

The neural network model i.e. BPNN as discussed in Section II for fault identification is constructed with $8-10-1/4$ layout i.e. 8 neurons input layer, 10 neurons hidden layer (one only) and $1/4$ neurons output layer. The targets of BPNN output layer for fault detection and classification under different fault scenarios are tabulated in Table I and II respectively.

TABLE II
TARGET OF BPNN OUTPUT LAYER FOR FAULT CLASSIFICATION

Fault Type	Network Outputs			
	A	B	C	G
No Fault	0	0	0	0
A-G	1	0	0	1
B-G	0	1	0	1
C-G	0	0	1	1
A-B-G	1	1	0	1
A-C-G	1	0	1	1
B-C-G	0	1	1	1
A-B	1	1	0	0
A-C	1	0	1	0
B-C	0	1	1	0
A-B-C	1	1	1	0
A-B-C-G	1	1	1	1

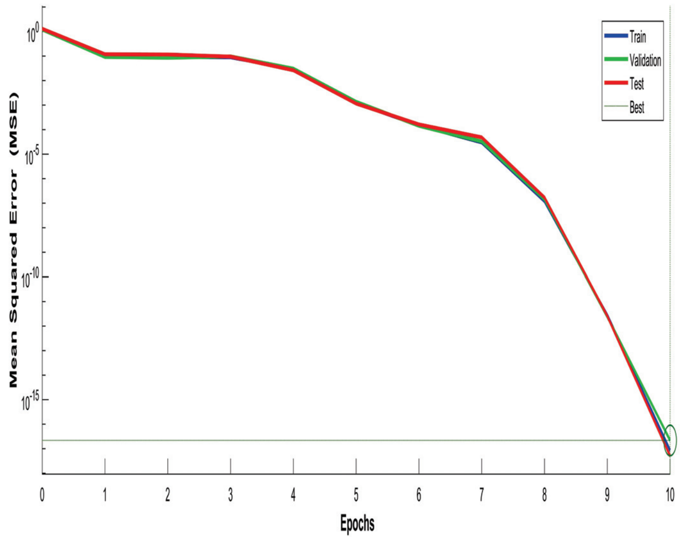

Fig. 4. Performance characteristics of data-sets for fault detection

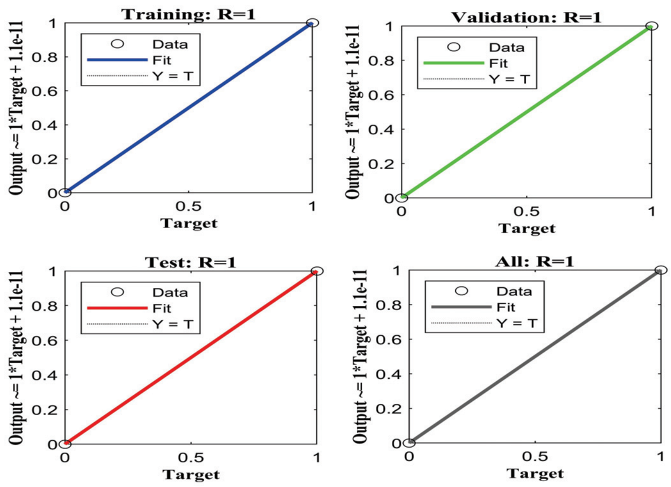

Fig. 5. Regression characteristics of data-sets for fault detection

To attain optimized weights with minimum Mean square error (MSE), the BPNN model is trained, tested and validated with fault voltages/current along with zero sequence volt-

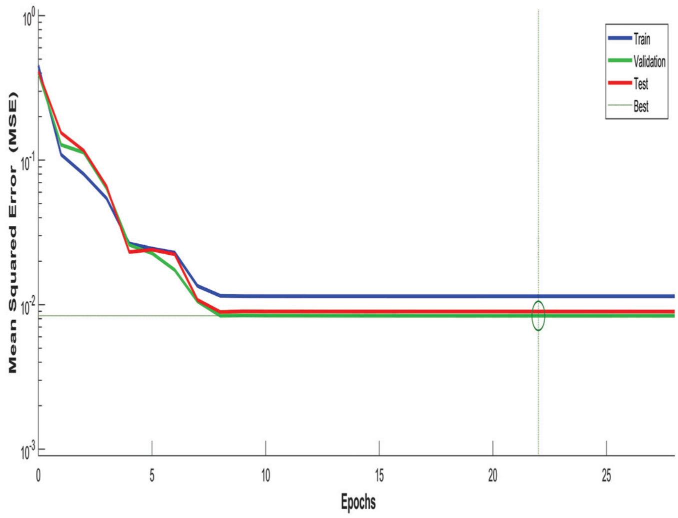

Fig. 6. Performance characteristics of data-sets for fault classification

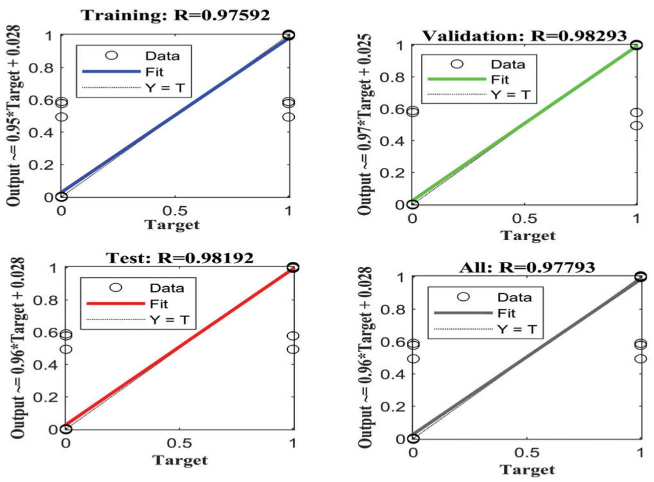

Fig. 7. Regression characteristics of data-sets for fault classification

age/current as inputs under various fault scenarios including low and high resistance faults on the transmission lines. The performance and regression characteristics of BPNN weights tuning during fault detection and classification are depicted in Fig. 4, Fig. 5 and Fig. 6, Fig. 7 respectively.

Based on above performance and regression characteristics, the efficacy of trained, tested and validated BPNN for fault detection and classification is observed to be 100% and 97.80% respectively. Thus, the simulation results reveal that the optimized weights for BPNN are attained with required MSE suitable to detect and classify the faults in interconnected transmission lines of the complex power system.

B. Performance of proposed fault identification methodology

The trained, validated and tested BPNN model as discussed in Subsection III A is simulated for various symmetrical and unsymmetrical faults on IEEE 14-bus test power system.

Various fault scenarios with low (0.1 Ω) and high (100 Ω) fault resistances is simulated on transmission line $1-5$. Further, the respective faulted three phase voltages/currents along with zero sequence voltage/current are fed to trained BPNN model. The detection performance of proposed BPNN under different fault scenarios with low (0.1 Ω) and high (100 Ω) fault resistances is depicted in Table III and Table IV

TABLE III
PERFORMANCE OF BPNN DURING FAULT DETECTION WITH LOW (0.1 Ω) FAULT RESISTANCE SCENARIOS

Fault Types	Network Inputs								Output
	V_a	V_b	V_c	I_a	I_b	I_c	V_z	I_z	
No Fault	4120.366	4120.304	4120.413	5.564738	5.564687	5.564833	6.57E-09	3.82E-11	0
A-G	4.044803	4424.027	4522.337	15.74242	5.332819	4.62094	2542.798	6.304387	1
B-G	4510.658	1.44821	4420.605	4.645003	15.74218	5.323108	2530.961	6.274794	1
C-G	4420.565	4510.552	1.520695	5.322895	4.644927	15.7424	2531.044	6.275001	1
A-B-G	1.547599	1.522865	4663.361	15.74201	15.7419	4.256004	2197.7	5.44857	1
A-C-G	1.513096	4663.281	1.638491	15.74217	4.255913	15.74227	2197.608	5.448341	1
B-C-G	4663.363	1.556661	1.504288	4.255896	15.74189	15.74231	2197.677	5.448515	1
A-B-C-G	1.574013	1.573991	1.573991	15.74194	15.74193	15.74195	1.58E-11	1.19E-11	1
A-B-C	1.574025	1.575292	1.575578	15.74194	15.74193	15.74195	1.65E-06	3.19E-05	1
A-B	2059.517	2059.623	4119.965	14.15452	13.66998	5.565907	0.390462	0.000968	1
A-C	2059.582	4119.885	2059.478	13.67022	5.565803	14.15477	0.390425	0.000968	1
B-C	4119.936	2059.503	2059.608	5.565854	14.15456	13.67014	0.390509	0.000968	1

TABLE IV
PERFORMANCE OF BPNN DURING FAULT DETECTION WITH HIGH (100 Ω) FAULT RESISTANCE SCENARIOS

Fault Types	Network Inputs								Output
	V_a	V_b	V_c	I_a	I_b	I_c	V_z	I_z	
No Fault	4120.366	4120.304	4120.413	5.564738	5.564687	5.564833	6.57E-09	3.82E-11	0
A-G	1009.299	4335.296	4415.106	13.24395	5.357107	4.826801	1912.453	4.74135	1
B-G	4415.041	1009.224	4335.343	4.82681	13.24376	5.357222	1912.388	4.74122	1
C-G	4335.287	4414.934	1009.259	5.356999	4.826721	13.24396	1912.464	4.741407	1
A-B-G	1095.556	1075.759	4543.838	13.0395	13.09809	4.544922	1716.027	4.254399	1
A-C-G	1075.78	4543.779	1095.571	13.09824	4.544872	13.03971	1715.973	4.254265	1
B-C-G	4663.363	1.556661	1.504288	4.255896	15.74189	15.7431	2197.677	5.448515	1
A-B-C-G	1139.412	1139.387	1139.41	12.92173	12.9216	12.92187	8.14E-09	1.80E-11	1
A-B-C	1139.412	1139.387	1139.41	12.92173	12.9216	12.92187	4.75E-09	7.83E-12	1
A-B	2258.903	2308.53	4119.985	11.75727	11.30159	5.565938	0.390392	0.000968	1
A-C	2308.324	4119.885	2258.859	11.30183	5.565803	11.75752	0.390374	0.000968	1
B-C	4119.941	2258.832	2308.381	5.565841	11.75727	11.30187	0.390387	0.000968	1

TABLE V
PERFORMANCE OF BPNN DURING FAULT CLASSIFICATION WITH LOW (0.1 Ω) FAULT RESISTANCE SCENARIOS

Fault Types	Network Inputs								Outputs			
	V_a	V_b	V_c	I_a	I_b	I_c	V_z	I_z	**A**	**B**	**C**	**G**
No Fault	4120.366	4120.304	4120.413	5.564738	5.564687	5.564833	6.57E-09	3.82E-11	0	0	0	0
A-G	4.044803	4424.027	4522.337	15.74242	5.332819	4.62094	2542.798	6.304387	1	0	0	1
B-G	4510.658	1.44821	4420.605	4.645003	15.74218	5.323108	2530.961	6.274794	0	1	0	1
C-G	4420.565	4510.552	1.520695	5.322895	4.644927	15.7424	2531.044	6.275001	0	0	1	1
A-B-G	1.547599	1.522865	4663.361	15.74201	15.7419	4.256004	2197.7	5.44857	1	1	0	1
A-C-G	1.513096	4663.281	1.638491	15.74217	4.255913	15.74227	2197.608	5.448341	1	0	1	1
B-C-G	4663.363	1.556661	1.504288	4.255896	15.74189	15.74231	2197.677	5.448515	0	1	1	1
A-B-C-G	1.574013	1.573991	1.573991	15.74194	15.74193	15.74195	1.58E-11	1.19E-11	1	1	1	1
A-B-C	1.574025	1.575292	1.575578	15.74194	15.74193	15.74195	1.65E-06	3.19E-05	1	1	1	0
A-B	2059.517	2059.623	4119.965	14.15452	13.66998	5.565907	0.390462	0.000968	1	1	0	0
A-C	2059.582	4119.885	2059.478	13.67022	5.565803	14.15477	0.390425	0.000968	1	0	1	0
B-C	4119.936	2059.503	2059.608	5.565854	14.15456	13.67014	0.390509	0.000968	0	1	1	0

TABLE VI
PERFORMANCE OF BPNN DURING FAULT CLASSIFICATION WITH HIGH (100 Ω) FAULT RESISTANCE SCENARIOS

Fault Types	Network Inputs								Outputs			
	V_a	V_b	V_c	I_a	I_b	I_c	V_z	I_z	**A**	**B**	**C**	**G**
No Fault	4120.366	4120.304	4120.413	5.564738	5.564687	5.564833	6.57E-09	3.82E-11	0	0	0	0
A-G	4.044803	4424.027	4522.337	15.74242	5.332819	4.62094	2542.798	6.304387	1	0	0	1
B-G	4510.658	1.44821	4420.605	4.645003	15.74218	5.323108	2530.961	6.274794	0	1	0	1
C-G	4420.565	4510.552	1.520695	5.322895	4.644927	15.7424	2531.044	6.275001	0	0	1	1
A-B-G	1.547599	1.522865	4663.361	15.74201	15.7419	4.256004	2197.7	5.44857	1	1	0	1
A-C-G	1.513096	4663.281	1.638491	15.74217	4.255913	15.74227	2197.608	5.448341	1	0	1	1
B-C-G	4663.363	1.556661	1.504288	4.255896	15.74189	15.74231	2197.677	5.448515	0	1	1	1
A-B-C-G	1.574013	1.573991	1.573991	15.74194	15.74193	15.74195	1.58E-11	1.19E-11	1	1	1	1
A-B-C	1.574025	1.575292	1.575578	15.74194	15.74193	15.74195	1.65E-06	3.19E-05	1	1	1	0
A-B	2059.517	2059.623	4119.965	14.15452	13.66998	5.565907	0.390462	0.000968	1	1	0	0
A-C	2059.582	4119.885	2059.478	13.67022	5.565803	14.15477	0.390425	0.000968	1	0	1	0
B-C	4119.936	2059.503	2059.608	5.565854	14.15456	13.67014	0.390509	0.000968	0	1	1	0

respectively. The simulation results reveal that the proposed BPNN with appropriate weights has accurately detected all symmetrical and unsymmetrical fault scenarios in the complex interconnected transmission lines.

Further, similar scenarios are simulated on transmission line $2-4$ to showcase the fault classification performance of proposed BPNN model. The classification performance of proposed BPNN under different fault scenarios with low ($0.1\ \Omega$) and high ($100\ \Omega$) fault resistances is depicted in Table V and Table VI respectively. The simulation results reveal that the proposed BPNN with appropriate weights has accurately classified most of the symmetrical and unsymmetrical fault scenarios in complex interconnected transmission lines.

Thus, the proposed fault detection and classification using artificial intelligence i.e. BPNN can be quite suitable for protection of complex interconnected transmission lines.

IV. CONCLUSION

The methodology for fault detection and classification in interconnected transmission lines by utilizing artificial intelligence i.e. Back Propagation Neural Network (BPNN) is proposed in this paper. The BPNN model trained, validated and tested with three voltages/currents along with zero sequence components through Levenberg-Marquardt algorithm and attained accuracy of 100% and 97.80% during fault detection and classification respectively. Various types of symmetrical and unsymmetrical faults are simulated on IEEE 14-bus power system to reveal the efficacy of proposed BPNN based fault identification strategy. The simulation results reveal that the proposed fault identification methodology has accurately detected and classified all types of transmission line fault events in the complex power system. Thus, the proposed artificial intelligence based fault identification concept can be quite suitable for protection of transmission lines in the electric grid.

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